Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $x=\cos\theta, y=\sin^3\theta$, then $\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2+y\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ at $\theta=\pi/2$ is

(A) 1

(B) 2

(C) -2

(D) -3

Q.2

If $f(x)=(1+x)^n$, then the value of $f(0)+f'(0)+\frac{1}{2!}f''(0)+\cdots+\frac{1}{n!}f^n(0)$ is

(A) n

(B) 2^n

(C) 2^{n-1}

(D) none of these

Q.3

If $y^2=P(x)$ is a polynomial of degree 3 , then $\frac{\mathrm{d}}{\mathrm{d}x}\left(y^3\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right)$ is equal to

(A) $P(x)+P''(x)$

(B) $P(x)P''(x)$

(C) $P(x)P'''(x)$

(D) a constant

Q.4

If $f(x)=x^{1/x}$ then $f''(e)$ is equal to

(A) $e^{1/(e-3)}$

(B) $e^{1/e}$

(C) $e^{1/(e-2)}$

(D) none of these

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Q.5

Let $f(x)$ be a polynomial of degree two which is positive for all $x \in \mathbf{R}$. If $g(x) = f(x) + f'(x) + f''(x) + xf'''(x) + x^2 f^{\text{IV}}(x)$, then for any real x

(A) $g(x) < 0$

(B) $g(x) > 0$

(C) $g(x) = 0$

(D) $g(x) \geq 0$

Q.6

If $y = \tan^{-1}\left(\frac{\log ex^{-2}}{\log ex^2}\right) + \tan^{-1}\left(\frac{3+2\log x}{1-6\log x}\right)$ then $\frac{d^2y}{dx^2}$ is

(A) 2

(B) 1

(C) 0

(D) x

Q.7

A twice differentiable function $f : \mathbf{R} \to \mathbf{R}$ satisfies $\sin x \cos y(f(2x+2y) - f(2x-2y)) = \cos x \sin y$
$(f(2x+2y) + f(2x-2y))$ for all $x, y \in \mathbf{R}$ then

(A) $f''(x) = f(x) = 0$

(B) $4f''(x) + f(x) = 0$

(C) $f''(x) + f(x) = 0$

(D) $4f''(x) - f(x) = 0$

Q.8

If $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 23 & 17 & 13 \\ 1 & 1 & 1 \end{vmatrix}$, $x \in \mathbf{R}$, then $\frac{d^2y}{dx^2} + y$ is equal to

(A) 4

(B) -10

(C) 0

(D) 6

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Q.9

If $\Delta_1 = \begin{vmatrix} x & b & b \\ a & x & b \\ a & a & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & b \\ a & x \end{vmatrix}$, then

(A) $\Delta_1 = 3\left(\Delta_2\right)^2$

(B) $\frac{d}{dx}\left(\Delta_1\right) = 3\Delta_2$

(C) $\frac{d}{dx}\left(\Delta_1\right) = 3{\Delta_2}^2$

(D) $\Delta_1 = 3\Delta_2^{3/2}$

Q.10

If $y = \frac{f(x)}{\phi(x)}$ and $z = \frac{f'(x)}{\phi'(x)}$, then $\frac{f^n}{f} - \frac{\phi^n}{\phi} + \frac{2(y-z)}{f\phi}\left(\phi^\eta\right)^2 =$

(A) $\frac{d^2y}{dx^2}$

(B) $\frac{1}{y}\frac{d^2y}{dx^2}$

(C) $y\frac{d^2y}{dx^2}$

(D) None of these

Q.11

If $x = \cos^7\theta$ and and $y = \sin\theta$, then $\frac{d^3x}{dy^3} =$

(A) $\frac{105}{4}\sin 4\theta$

(B) $\frac{105}{2}\sin 2\theta$

(C) $\frac{105}{4}\cos 4\theta$

(D) None of these

Q.12

If $f(x) = x^3 + x^2 f'(1) + xf''(2) + f''(3)$ for all $x \in R$, then which of the following is false?

(A) $f(0) + f(2) = f(1)$

(B) $f(0) + f(3) = 0$

(C) $f(1) + f(3) = f(2)$

(D) $f(\mathrm{I}) + f(3) = f(0)$

Q.13

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Suppose the function $f(x) - f(2x)$ has the derivative 5 at $x = 1$ and derivative 7 at $x = 2$. The derivative of the function $f(x) - f(4x)$ at $x = 1$ has the value equal to

(A) 19

(B) 9

(C) 17

(D) 14

Q.14

If $f(x) = \frac{1-\cos x}{1-\sin x}$, then $f'\left(\frac{\pi}{2}\right)$ is equal to

(A) 1

(B) 0

(C) ∞

(D) does not exist

Q.15

If $f,\ \mathrm{g}, h$ are differentiable functions of x and $\Delta = \begin{vmatrix} f & g & h \\ (xf)' & (xg)^1 & (xh)' \\ (x^2f)' & (x^2g)' & (x^2h) \end{vmatrix}$ then Δ' (the derivative of Δ w.r.t. x) in given by

(A) $\begin{vmatrix} f' & g' & h' \\ f & g & h \\ (x^3f'')' & (x^3g'')' & (x^3h) \end{vmatrix}$

(B) $\begin{vmatrix} f & g & h \\ f' & g' & h' \\ (x^2f'')' & (x^2g'')' & (x^2h')' \end{vmatrix}$

(C) $\begin{vmatrix} f & g & h \\ f' & g^\dagger & h' \\ (x^3f'')^1 & (x^3g'')' & (x^3h')' \end{vmatrix}$

(D) None of these

Answers & Solutions

Q.1 Answer:

-3

Solution:

$\frac{dx}{d\theta} = -\sin\theta,\ \frac{dy}{d\theta} = 3\sin^2\theta\cos\theta \Rightarrow$
$\frac{dy}{dx} = -3\sin\theta\cos\theta$. **Differentiating again, we have**

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$$\frac{d^2y}{dx^2} = -3\cos^2\theta\frac{d\theta}{dx} = \frac{3\cos 2\theta}{\sin\theta}$$

$$\left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2} = 9\sin^2\theta\cos^2\theta + \sin^3\theta\frac{3\cos 2\theta}{\sin\theta}$$

$$= 9\sin^2\theta\cos^2\theta + 3\sin^2\theta\cos 2\theta$$

So the value of the expression on L.H.S. at $\theta = \pi/2$ is -3.

Q.2 Answer:

2^n

Solution:

$f(0) = 1, f'(x) = n(1+x)^{n-1},$
$f''(x) = n(n-1)(1+x)^{n-2} \cdots f^n(x) = n(n-1)\cdots 1(1+$
$x)^{n-n}$
So $f'(0) = n, f''(0) = n(n-1), \ldots f^n(0) = n!.$
$\therefore$ the given expression is equals to $1 + n + \frac{n(n-1)}{2!} + \cdots \frac{n!}{n!} = 2^n$

Q.3 Answer:

$P(x)P'''(x)$

Solution:

From $y^2 = P(x)$**, we have** $2yy_1 = P'(x)$ **i.e.** $2y_1 = P'(x)/y$
$\Rightarrow \quad 2y_2 = \frac{yP''(x) - P'(x)y_1}{y^2} = \frac{yP''(x) - P'(x)P'(x)/2y}{y^2}$
$\Rightarrow \frac{2y^2P''(x) - \left(P'(x)\right)^2}{2y^3}$
$\Rightarrow 2y_2y^3 = \frac{1}{2}\left[2P(x)P''(x) - \left(P'(x)\right)^2\right]$
$\Rightarrow \quad 2\frac{d}{dx}\left(y^3\frac{d^2y}{dx^2}\right) = \frac{1}{2}\left[2\left\{P'(x)P''(x) + P(x)P'''(x)\right\} - 2P'(x)P''(x)\right]$
$= \frac{1}{2}2P(x)P'''(x) = P(x)P'''(x)$

Q.4 Answer:

none of these

Solution:

$\log f(x) = \frac{1}{x}\log x$**. Differentiating both sides we have**

$$\frac{f'(x)}{f(x)} = \frac{1}{x^2} - \frac{\log x}{x^2}$$

$$\Rightarrow \qquad f'(x) = f(x)\frac{1 - \log x}{x^2}$$

Differentiating again both the sides, we have

$f''(x) = f'(x)\frac{1-\log x}{x^2} + f(x)\left[\frac{x^2(-1/x) - (1-\log x)2x}{x^4}\right]$
thus $f''(e) = -\frac{f(e)}{e^3} = -e^{(1/e)-3}.$

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Q.5 Answer:

$g(x) > 0$

Solution:

Let $f(x) = ax^2 + bx + c$. As $f(x) > 0$ for all $x \in \mathbf{R}$, we must have, $a > 0$ and $b^2 - 4ac < 0$

Now

$$g(x) = ax^2 + bx + c + (2ax + b) + 2a + x.0 + x^2 \cdot 0$$
$$= ax^2 + (b + 2a)x + b + c + 2a$$
$$\text{Discriminant of } g(x) = (b + 2a)^2 - 4a(b + c + 2a)$$
$$= -4a^2 + \left(b^2 - 4ac\right) < 0$$

Thus $g(x) > 0$ for all $x \in \mathbf{R}$.

Q.6 Answer:

0

Solution:

$$y = \tan^{-1}\left(\frac{\log e - 2\log x}{\log e + 2\log x}\right) + \tan^{-1}\left(\frac{3 + 2\log x}{1 - 6\log x}\right)$$
$$= \tan^{-1}\left(\frac{1 - 2\log x}{1 + 2\log x}\right) + \tan^{-1}\left(\frac{3 + 2\log x}{1 - 6\log x}\right)$$
$$= \frac{\pi}{4} - \tan^{-1}(2\log x) + \tan^{-1} 3 + \tan^{-1}(2\log x)$$
$$= \frac{\pi}{4} + \tan^{-1} 3$$
$$\text{Thus} \quad \frac{dy}{dx} = 0 \Rightarrow \frac{d^2y}{dx^2} = 0$$

Q.7 Answer:

$4f''(x) + f(x) = 0$

Solution:

$$f(2x + 2y)(\sin x \cos y - \cos x \sin y)$$
$$\Rightarrow \quad f(2x + 2y)\sin(x - y) = f(2x - 2y)\sin(x + y)$$
$$\Rightarrow \quad \frac{f(2(x+y))}{\sin(x+y)} = \frac{f(2x-2y)}{\sin(x-y)}$$

Putting $2(x + y) = u, 2(x - y) = v$

$$\Rightarrow \quad \frac{f(u)}{\sin\frac{u}{2}} = \frac{f(v)}{\sin\frac{v}{2}} = k \text{ for all } u, v \in \mathbf{R}$$
$$\Rightarrow \quad f(u) = k\sin\frac{u}{2}$$
$$\Rightarrow \quad f'(u) = \frac{k}{2}\cos\frac{u}{2} \text{ and } f''(u) = \frac{-k}{4}\sin\frac{u}{2}$$
$$\Rightarrow \quad 4f''(u) = -k\sin\frac{u}{2} = -f(u)$$
$$\Rightarrow \quad 4f''(u) + f(u) = 0$$

Q.8 Answer:

6

Solution:

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Using $C_2 \to C_2 - C_1, C_3 \to C_3 - C_1$, we get

$$y(x) = \begin{vmatrix} \sin x & \cos x - \sin x & \cos x + 1 \\ 23 & -6 & -10 \\ 1 & 0 & 0 \end{vmatrix}$$

$= -10(\cos x - \sin x) + 6\cos x + 6$

$= 10\sin x - 4\cos x + 6$

$\frac{dy}{dx} = 10\cos x + 4\sin x$

$\frac{d^2y}{dx^2} = -10\sin x + 4\cos x = -y + 6$

$\Rightarrow \frac{d^2y}{dx^2} + y = 6$

Q.9 Answer:

$\frac{d}{dx}(\Delta_1) = 3\Delta_2$

Solution:

We have

$$\frac{d\Delta_1}{dx} = \begin{vmatrix} 1 & b & b \\ 0 & x & b \\ 0 & a & x \end{vmatrix} + \begin{vmatrix} x & 0 & b \\ a & 1 & b \\ a & 0 & x \end{vmatrix} + \begin{vmatrix} x & b & 0 \\ a & x & 0 \\ a & b & 1 \end{vmatrix}$$

$$= \begin{vmatrix} x & b \\ a & x \end{vmatrix} + \begin{vmatrix} x & b \\ a & x \end{vmatrix} + \begin{vmatrix} x & b \\ a & x \end{vmatrix}$$

$= 3\Delta_2.$

Q.10 Answer:

$\frac{1}{y}\frac{d^2y}{dx^2}$

Solution:

$\frac{dy}{dx} = \frac{f'\phi - f\phi'}{\phi^2}$

$\frac{d^2y}{dt^2} = \frac{(f''\phi - f\phi^n)\phi^2 - (f'\phi - f\phi')2\phi\phi'}{\phi^4}$

$\therefore \frac{1}{y}\frac{d^2y}{dx^2} = \frac{f''\phi - f\phi^\circ}{f\phi} - \frac{2(f'\phi - f\phi)}{f\phi^2\phi'}(\phi')^2$

$= \frac{f''}{f} - \frac{\phi^n}{\phi} + \frac{2(y-z)}{f\phi}(\phi)^2$

Q.11 Answer:

$\frac{105}{4}\sin 4\theta$

Solution:

We have, $x = \cos^7\theta$ **and** $y = \sin\theta$

Differentiating w.r.t. θ**, we get**

$\frac{dx}{d\theta} = -7\cos^6\theta\sin\theta$

and, $\frac{dy}{d\theta} = \cos\theta$ **thus, we have,** $\frac{dx}{dy} = -7\cos^5\theta\sin\theta$

and, $\frac{d^2x}{dy^2} = \frac{d}{dy}\left(\frac{dx}{dy}\right) = \frac{d}{d\theta}\left(\frac{dx}{dy}\right)\cdot\frac{d\theta}{dy}$

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$= \left(35\cos^4\theta\sin^2\theta - 7\cos^6\theta\right)\frac{1}{\cos\theta} = 35\cos^3\theta\sin^2\theta - 7\cos^5\theta$

$= 35\cos^3\theta\left(1-\cos^2\theta\right) - 7\cos^5\theta = 35\cos^3\theta - 42\cos^5\theta$

and, $\frac{d^3x}{dy^3} = \frac{d}{dy}\left(\frac{d^2x}{dy^2}\right) = \frac{d}{d\theta}\left(\frac{d^2x}{dy^2}\right)\cdot\frac{d\theta}{dy}$

$= \left(-105\cos^2\theta\sin\theta + 210\cos^4\theta\sin\theta\right)\frac{1}{\cos\theta} = 105\sin\theta\cos\theta\left(2\cos^2\theta - 1\right)$

$= \frac{105}{2}\sin 2\theta\cos 2\theta = \frac{105}{4}\sin 4\theta$

Q.12 Answer:

$f(\mathrm{I}) + f(3) = f(0)$

Solution:

$f(x) = x^3 + x^2 f'(1) + xf''(2) + f'''(3)$

$\therefore f'(x) = 3x^2 + 2xf'(1) + f''(2)$

$\Rightarrow \quad f''(x) = 6x + 2f'(1) \quad \Rightarrow \quad f''(x) = 6$

Now, $f'(1) = 3 + 2f'(1) + f''(2)$

$\Rightarrow \quad f''(2) + f'(1) + 3 = 0 \quad \ldots(1)$

Again, $f''(2) = 12 + 2f'(1)$

$\Rightarrow \quad f''(2) - 2f'(1) - 12 = 0 \quad \ldots.(2)$

Again, $f'''(3) = 6$

$\ldots.(3)$

From (1) and (2)

$2f''(2) + f''(2) - 6 = 0 \Rightarrow f''(2) = 2 \quad \ldots(4)$

$\therefore$ **(1) gives** $f'(1) + 2 + 3 = 0$

$\Rightarrow f'(1) = -5$

$\therefore \quad f(x) = x^3 - 5x^2 + 2x + 6$

$\therefore f(0) = 6, \quad f(1) = 1 - 5 + 2 + 6 = 4$

$f(2) = 8 - 20 + 4 + 6 = -2$

$f(3) = 27 - 45 + 6 + 6 = -6 \quad f(0) + f(2) = 6 - 2 = f(1)$

$\therefore$ **(a) is true**

$f(0) + f(3) = 6 - 6 = 0$

$f(1) + f(3) = (1 - 5 + 2 + 6) - 6$

$= -2 = f(2)$

$f(1) + f(3) = -2 \neq f(0) \quad [\because f(0) = 6]$

$\therefore$ **(d) is false**

Q.13 Answer:

19

Solution:

$y = f(x) - f(2x) \Rightarrow y^1 = f'(x) - 2f'(2x)$

$\Rightarrow y'(1) = f'(1) - 2f'(2) = 5,$ **and** $\quad \ldots\ldots(1)$

$y'(2) = f'(2) - 2f'(4) = 7 \quad \ldots\ldots(2)$

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Now let $y = f(x) - f(4x)$

$\Rightarrow y^{\dagger} = f'(x) - 4f'(4x)$

$\Rightarrow y(1) = f'(1) - 4f'(4) \quad \ldots\ldots(3)$

Substituting the value of $f'(2) = 7 + 2f'(4)$ **in** (1)**, we get**

$f'(1) - 2(7 + 2f'(4)) = 5$

$f'(1) - 4f'(4) = 19$

Q.14 Answer:

does not exist

Solution:

$f(x) = \frac{(1-\sin x)(\sin x)-(1-\cos x)(-\cos x)}{(1-\sin x)^2}$

$= \frac{\sin x-\sin^2 x+\cos x-\cos^2 x}{(1-\sin x)^2} = \frac{\sin x+\cos x-1}{(1-\sin x)^2}$

$\therefore f'\left(\frac{\pi}{2}\right) = \frac{1+0-1}{(1-1)^2} = \frac{0}{0}$

Therefore, $f'\left(\frac{\pi}{2}\right)$ **does not exist.**

Q.15 Answer:

$$\begin{vmatrix} f & g & h \\ f' & g^{\dagger} & h' \\ \left(x^3 f''\right)^1 & \left(x^3 g''\right)' & \left(x^3 h'\right)' \end{vmatrix}$$

Solution:

We have,

$$\Delta = \begin{vmatrix} f & g & h \\ x' + f & xg' + g & xh' + h \\ x^2 f'' + 4x'' + 2f & x^2 g'' + 4xg' + 2g & x^2 h'' + 4xh^4 + 2h \end{vmatrix}$$

$$= \begin{vmatrix} f & g & h \\ xf' & xg^1 & xh^1 \\ x^2 f^n + 4yf' & x^2 g^n + 4xg' & x^2 h^n + 4xh \end{vmatrix}$$

[Applying $R_2 \to R_2 - R_1$ **and** $R_3 \to R_3 - 2R_1$**]**

$$= \begin{vmatrix} f & g & h \\ xf' & xg' & xh' \\ x^2 f'' & x^2 g'' & x^2 h^2 \end{vmatrix}$$ **[Applying** $R_3 \to R_3 - 4R_2$**]**

$$= x^3 \begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h^2 \end{vmatrix} = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ x^3 f'' & x^3 g'' & x^3 h' \end{vmatrix}$$

$$\therefore \Delta' = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ \left(x^3 f''\right)' & \left(x^3 g''\right)' & \left(x^3 h'\right)' \end{vmatrix}$$

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